

ATPL EASA STUDY GUIDE | 032 - PERFORMANCE

Chapter 4: Class B En-Route & Factorised Landing Performance

Syllabus Source: CAE Oxford (pp. 115-148) | Pass Target: $\geq 90\%$ AviationExam

1. CRUISE PERFORMANCE: RANGE VS ENDURANCE

Cruising efficiency for propeller aircraft depends on matching engine brake horsepower to airframe drag curves:

- Maximum Range Cruise Speed (V_{MRC}): Flown at the speed of minimum total drag (V_{md} / tangent point on drag curve). Provides the maximum nautical miles per kilogram of fuel burned (NMPK).
- Maximum Endurance Speed (V_{min_power}): Flown at the speed of minimum power required ($V_{mp} \sim 0.76 \times V_{md}$). Provides the maximum flight time (hours) in the air per kilogram of fuel burned. Used for holding patterns!

2. CLASS B COMMERCIAL LANDING SAFETY FACTORS

Under EASA AIR-OPS (CAT.POL.A.330), landing distance on a commercial flight must incorporate legal safety factors:

- The 1.43 Landing Factor: The Landing Distance Available (LDA) at the destination must be at least 1.43 times the actual measured landing distance from 50 ft: Factored Landing Distance = Actual Landing Distance $\times$ 1.43.
- Grass Runway Corrections: Dry grass up to 20 cm: Multiply landing distance by 1.20 (+20%). Wet grass: Multiply by 1.30 (+30%).
- Slope Corrections: Downslope runway: Increases landing distance by +5% per 1% downhill slope!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Class B Landing Distance Factor

Problem: Flight manual demonstration distance from 50 ft to full stop is 500 m.

Landing Distance Available (LDA) is 850 m. Step 1: Calculate Dry Runway Required

Landing Distance (RLD): - $RLD_{dry} = 500 \text{ m} \times 1.43 = 715 \text{ meters}$ (must stop within 70%

of LDA: $1/0.70 = 1.43$). - Check: $715 \text{ m} \leq 850 \text{ m} \rightarrow$ Dry landing is LEGAL! Step 2:

Calculate Wet Runway Required Landing Distance: - $RLD_{wet} = RLD_{dry} \times 1.15 = 715 \text{ m} \times$

$1.15 = 822.25 \text{ meters}$. - Check: $822.25 \text{ m} \leq 850 \text{ m} \rightarrow$ Wet landing is LEGAL!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Factored Landing Distance on Wet Grass Runway

SCENARIO (Commercial Class B Landing Problem): A twin-propeller air-taxi is planning to land at an unpaved airfield: - Unfactored (Flight Manual) Landing Distance from 50 ft on dry asphalt = 500 meters - Destination runway: Wet grass surface (+30% correction) - Runway slope: 2.0% downhill slope (+5% per 1% slope = +10% correction) - EASA commercial operation factor: 1.43 QUESTION: What is the MINIMUM Landing Distance Available (LDA) required to land legally? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Apply the Runway Surface Correction (Wet Grass = x 1.30): - Distance after grass factor = $500 \text{ m} \times 1.30 = 650 \text{ meters}$. Step 2: Apply the Downslope Correction (2% downhill x 5% = +10% -> x 1.10): - Distance after slope factor = $650 \text{ m} \times 1.10 = 715 \text{ meters}$. Step 3: Apply the mandatory EASA Commercial Safety Factor (x 1.43): - Factored Required LDA = $715 \text{ m} \times 1.43 = 1,022.45 \text{ meters}$ -> 1,023 meters! FINAL ANSWER: The runway must have a certified Landing Distance Available (LDA) of AT LEAST 1,023 meters! (More than double the basic 500 m manual distance!).

[!] EXAM TRAP (AVIATIONEXAM) - The Sequential Multiplication Trap

- Always apply factors sequentially (Surface -> Slope -> Regulatory factor). - Notice how a 500 m basic distance turns into 1,023 m! In ATPL exams, failing to apply the 1.43 commercial factor is the most common student error!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under EASA regulations, what is the regulatory multiplication factor applied to the landing distance of a C

- [A]: 1.15
- [B]: 1.43 (meaning required LDA must be at least 143% of actual distance)
- [C]: 1.67
- [D]: 2.00

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under CAT.POL.A.330 for Class B aircraft, the landing distance from 50 ft must be multiplied by 1.43 (equivalent to using no more than 70% of the available runway).

Q2: At what airspeed does a piston-propeller aircraft achieve MAXIMUM ENDURANCE (maximum time in the ai

- [A]: At the speed for minimum power required ($V_{mp} \sim 0.76 V_{md}$)
- [B]: At the speed for minimum total drag (V_{md})
- [C]: At V_{no}
- [D]: At stall speed

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [A]. Piston engine fuel burn is proportional to brake horsepower. Flying at the bottom of the Power Required curve (V_{mp}) burns the minimum fuel per hour, maximizing endurance.

Q3: How does a DOWNHILL runway slope affect landing distance?

- [A]: Decreases landing distance
- [B]: INCREASES landing distance (approx +5% per 1% of downhill slope)
- [C]: Has zero effect on landing roll

- [D]: Increases tire wear only

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. A downhill slope reduces the deceleration component of gravity, requiring more wheel braking and lengthening landing distance.